

Simple portfolios also provide strong benchmarks to measure the value-added of more complicated statistical and economic models.

The simplest strategy—an equally weighted portfolio—turns out to be one of the best performers, as we shall now see.

5. Special Mean-Variance Portfolios

In this section I run a horse race between several portfolio strategies, each of which is a special case of the full mean-variance strategy. Diversification is common to all the strategies, but they build a diversified portfolio in different ways. This leads to very different performance.

5.1. HORSERACE

I take four asset classes—U.S. government bonds (Barcap U.S. Treasury), U.S. corporate bonds (Barcap U.S. Credit), U.S. stocks (S&P 500), and international stocks (MSCI EAFE)—and track the performance of various portfolios from January 1978 to December 2011. The data are sampled monthly. The strategies implemented at time t are estimated using data over the past five years, $t-60$ to t . The first portfolios are formed at the end of January 1978 using data from January 1973 to January 1978. The portfolios are held for one month, and then new portfolios are formed at the end of the month. I use one-month T-bills as the risk-free rate. In constructing the portfolios, I restrict shorting down to -100% on each asset class.

Using short, rolling samples opens me up to the criticisms of the previous section. I do this deliberately because it highlights some of the pitfalls of (fairly) unconstrained mean-variance approaches. Consequently, it allows us to understand why some special cases of mean-variance perform well and others badly.

I run a horserace between:

Mean-Variance Weights where the weights are chosen to maximize the Sharpe ratio.

Market Weights, which are given by market capitalizations of each index.

Diversity Weights, which are (power) transformations of market weights recommended by Fernholz, Garvy, and Hannon (1998).

Equal Weights, or the $1/N$ rule, which simply holds one-quarter in each asset class. Duchin and Levy (2009) call this strategy the “Talmudic rule” since the Babylonian Talmud recommended this strategy approximately 1,500 years ago: “A man should always place his money, one third in land, a third in merchandise, and keep a third in hand.”